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LEARNER GUIDE

For

Data Analytics with R

Course Code: C182

Conducted by
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DOCUMENT VERSION CONTROL RECORD

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How to Use This Guide

The PowerPoint is intentionally visual and concept-led. This Learner Guide carries the detailed procedures for ten connected R labs. Each lab folder is self-contained: run its R script, inspect generated outputs and complete the Excel intake/control/evidence workbook.

LMS/TMS: https://lms-tms.tertiaryinfotech.com/

GitHub courseware:https://github.com/tertiarycourses/C182-Data-Analytics-with-R

Learning Outcomes

- LO1: Prepare and transform data for analysis using R.
- LO2: Summarise data with descriptive statistics, figures, graphs and tables.
- LO3: Perform quantitative data analysis and interpret the findings.
- LO4: Perform qualitative data analysis and determine significance or effect.
- LO5: Create effective data visualisations in R and interpret research findings.

R Setup and Evidence Conventions

- R 4.2 or later and RStudio Desktop are recommended.
- The scripts run with base R; optional ggplot2/wordcloud enhancements activate only when installed.
- Do not edit the supplied source CSV files. Derived CSV, image and session evidence belongs in outputs/.
- Keep the Excel workbook beside the script and use it for the scenario, independent controls, evidence and run log.
- Treat all healthcare records as synthetic training data. Do not make clinical or population claims from them.

Topic 01: Data Preparation and Transformation

Research-data workflow · import/export · tidy structure · cleaning · joins · transformation

Research data becomes evidence through a controlled workflow

- **Frame:** Define the research question, population, outcome and unit of observation.
- **Acquire:** Record source, owner, period, consent and extraction conditions.
- **Prepare:** Type, clean, transform and join without destroying source evidence.
- **Validate:** Reconcile records and totals before analysis or visualisation.

R analysis packages form a coherent toolkit

- **Base R:** Import, inspect, reshape, model and plot with the core language.
- **tidyverse:** A consistent grammar for import, manipulation and visualisation.
- **readxl:** Reads Excel worksheets while preserving typed rectangular data.
- **ggplot2:** Builds layered graphics from data, aesthetics and geoms.

Tidy data makes transformations composable

- **Variable:** Each measured attribute occupies one column.
- **Observation:** Each unit or event occupies one row.
- **Value:** Each cell contains one atomic value.
- **Table:** Each observational unit forms its own analytical table.

Import is a data-contract decision

- **Delimiter:** Confirm comma, tab or other field boundaries.
- **Header:** Use stable unique names that survive refresh.
- **Type:** Declare dates, identifiers, numbers and categories deliberately.
- **Locale:** Check date order, decimal marks and character encoding.

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CSV and Excel serve different purposes

- **CSV:** Portable plain text with one table and no formulas or styles.
- **Excel:** Multiple sheets, formulas and formatted control/evidence surfaces.
- **Risk:** Spreadsheet display can hide underlying types or errors.
- **Control:** Keep an analysis-ready CSV beside the learner evidence workbook.

Select columns to define analytical scope

- **Relevance:** Keep variables required by the question and validation controls.
- **Privacy:** Exclude identifiers or sensitive fields that are not necessary.
- **Order:** Place keys, dimensions and measures in a reviewable sequence.
- **Lineage:** Document fields removed or renamed and why.

Filter rows with explicit inclusion rules

- **Condition:** Write the population rule before seeing the result.
- **Boolean logic:** Use AND, OR and NOT with deliberate parentheses.
- **Missing values:** NA comparisons require explicit is.na() handling.
- **Evidence:** Record input, excluded and output row counts.

Logical operators define the population

- **AND:** Every condition must be true for a row to remain.
- **OR:** At least one condition is sufficient.
- **NOT:** Excludes a defined condition rather than an informal exception.
- **Audit:** Test rules on known boundary records before full use.

Missingness is a pattern, not merely a nuisance

- **Unknown:** The value was not observed or recorded.
- **Not applicable:** The variable has no meaning for this observation.
- **Invalid:** A value fails type, range or relationship rules.
- **Decision:** Exclude, impute or retain with a documented rationale.

Mutate creates derived analytical variables

- **Formula:** Express the relationship in visible code, such as BMI = weight/height².
- **Unit:** Check that source units make the result meaningful.
- **Boundary:** Test zero, negative, missing and extreme inputs.
- **Trace:** Keep the original columns beside the derived value.

Dates turn periods into ordered evidence

- **Source:** Year and week fields need a documented calendar convention.
- **Parse:** Convert strings to true dates instead of plotting labels.
- **Validate:** Reject impossible or ambiguous dates visibly.
- **Use:** True dates enable sorting, intervals, trends and seasonality.

Joins enrich data—and can corrupt it silently

- **Left join:** Preserves every analytical record while adding reference attributes.
- **Inner join:** Keeps only matched records and can hide missing keys.
- **Many-to-many:** Duplicates records and inflates totals when keys are not unique.
- **Control:** Check key uniqueness, unmatched keys and row preservation.

Join type follows the analytical question

- **Left:** Keep all observations from the primary dataset.
- **Inner:** Retain only keys present in both tables when justified.
- **Full:** Surface every key from both sources for reconciliation.
- **Anti:** Isolate unmatched records as a quality exception.

Reshaping changes structure, not meaning

- **Long:** One measurement column plus key/value variables supports grouping and plotting.
- **Wide:** Repeated categories occupy columns for comparison or delivery.
- **Identifier:** Keys must remain stable through every pivot.

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- **Reconcile:** Totals and distinct observations should remain explainable.

Aggregation changes the unit of observation

- **Group:** Define the dimensions that create each summary row.
- **Measure:** Choose sum, count, mean or another statistic intentionally.
- **Denominator:** Retain n and exposure when comparing groups.
- **Risk:** Aggregates can conceal within-group variation and missingness.

Pipes make the analytical story readable

- **Sequence:** Each operation receives the result of the previous stage.
- **Names:** Assign meaningful intermediate outputs at review boundaries.
- **Debug:** Inspect after import, join, filter and aggregation gates.
- **Reproducibility:** The same source and code should recreate the result.

Reconciliation catches silent loss

- **Rows:** Input = retained + documented exclusions.
- **Keys:** Distinct business keys remain unique where required.
- **Totals:** Material measures agree before and after transformation.
- **Exceptions:** Unmatched, invalid or missing records are countable and owned.

A release-ready dataset carries evidence

- **Source:** Origin, extraction date and owner.
- **Code:** Versioned script and package/session information.
- **Controls:** Row, key, total and exception checks.
- **Limitations:** Known missingness, bias, transformations and allowed use.

Lab 1: Prepare and Transform Dengue Surveillance Data

Alignment	LO1
R script	lab-01-data-preparation.R
Excel evidence	lab-01-data-preparation.xlsx
Expected output	A reproducible clean dengue time-series dataset, transformation log and reconciliation evidence.
Tools	R, read.csv, dplyr-compatible workflow, date construction, write.csv

Goal. Import weekly dengue surveillance data, inspect types and missingness, filter a defined population, construct dates and export a controlled analysis table.

Before you begin

- Work from the numbered lab folder and keep its script, workbook and data subfolder together.
- Use the supplied synthetic/legacy training data only; it is not clinical evidence.
- Open the Excel workbook first and read the scenario, definitions and control expectations.
- Run the R script from the lab folder so relative data/ and outputs/ paths resolve correctly.
- Preserve the source CSV files. Put generated evidence only in outputs/ and the Excel Evidence sheet.

Detailed procedure

1	**Open the lab folder and review the scenario, data dictionary and control totals in the Excel workbook.** Open lab-01-data-preparation.xlsx and data/dengue.csv

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2	**Set the R working directory to this lab folder or open the folder as an RStudio Project.** Use Session > Set Working Directory > To Source File Location
3	**Run the script section that imports the CSV without changing the source file.** source('lab-01-data-preparation.R', echo = TRUE)
4	**Inspect column names, types, record count, distinct case types and missing values.** Review str(), summary(), table() and colSums(is.na()) output
5	**Keep complete Dengue records and construct a Monday date from year and epidemiological week.** Verify clean_dengue contains year, eweek, type_dengue, number and date
6	**Reconcile input, excluded and output rows and review the exported transformation log.** Open outputs/lab-01-transformation-log.csv
7	**Inspect the generated time-series plot and record one defensible observation in the Evidence sheet.** Open outputs/lab-01-dengue-trend.png
8	**Save the workbook evidence and retain the R console output.** Complete the Evidence and Reflection sheets

Troubleshooting

- File not found: confirm the script is running from the lab folder and the named CSV exists under data/.
- Package not installed: the supplied scripts use base-R fallbacks where possible; read the console message before installing anything.
- Unexpected factor or text values: inspect unique(), trim whitespace and compare with the workbook data dictionary.
- Plot file is blank or locked: close any open PNG, rerun the script and confirm dev.off() completed.
- Control does not reconcile: stop interpretation, identify the transformation stage where rows or totals diverged, and document the exception.

Evidence and acceptance

- Retain the runnable R script and full console output.
- Retain generated CSV/PNG/session evidence from outputs/.
- Complete the Excel Control, Evidence and Run Log sheets.
- State the finding, magnitude/effect, uncertainty, decision relevance and limitation in your own words.

The script runs without error, output rows reconcile to input less documented exclusions, dates are valid, and the clean CSV and plot are created.

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Lab 2: Join, Reshape and Aggregate Vaccination Data

Alignment	LO1
R script	lab-02-join-reshape.R
Excel evidence	lab-02-join-reshape.xlsx
Expected output	A joined long-form vaccination dataset, unmatched-key report, annual summary and transformation audit.
Tools	R, merge, reshape, aggregate, key validation, reconciliation

Goal. Standardise vaccination and clinic reference data, join on controlled keys, reshape between wide and long forms and produce an auditable annual summary.

Before you begin

- Work from the numbered lab folder and keep its script, workbook and data subfolder together.
- Use the supplied synthetic/legacy training data only; it is not clinical evidence.
- Open the Excel workbook first and read the scenario, definitions and control expectations.
- Run the R script from the lab folder so relative data/ and outputs/ paths resolve correctly.
- Preserve the source CSV files. Put generated evidence only in outputs/ and the Excel Evidence sheet.

Detailed procedure

1	**Review the vaccination and clinic reference tables plus their keys in the workbook.** Open lab-02-join-reshape.xlsx and data/*.csv
2	**Run the script and inspect uniqueness of vaccination_type in the clinic reference.** source('lab-02-join-reshape.R', echo = TRUE)
3	**Standardise text keys by trimming whitespace and applying consistent case.** Inspect the standardise_key() function and key audit
4	**Join the vaccination observations to the clinic reference using a left join.** Verify the joined row count equals the vaccination input row count
5	**Review and resolve any unmatched keys before interpreting the summary.** Open outputs/lab-02-unmatched-keys.csv
6	**Create annual totals by vaccine group and a wide comparison table.** Open outputs/lab-02-annual-summary.csv and lab-02-wide-summary.csv
7	**Check totals before and after reshaping and record the result in the Control sheet.** Input doses must equal grouped-summary doses

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8	**Write one limitation about aggregation or key quality in the Evidence sheet.** Complete Evidence!B6:B8

Troubleshooting

- File not found: confirm the script is running from the lab folder and the named CSV exists under data/.
- Package not installed: the supplied scripts use base-R fallbacks where possible; read the console message before installing anything.
- Unexpected factor or text values: inspect unique(), trim whitespace and compare with the workbook data dictionary.
- Plot file is blank or locked: close any open PNG, rerun the script and confirm dev.off() completed.
- Control does not reconcile: stop interpretation, identify the transformation stage where rows or totals diverged, and document the exception.

Evidence and acceptance

- Retain the runnable R script and full console output.
- Retain generated CSV/PNG/session evidence from outputs/.
- Complete the Excel Control, Evidence and Run Log sheets.
- State the finding, magnitude/effect, uncertainty, decision relevance and limitation in your own words.

The left join preserves all source rows, unmatched keys are zero or documented, and total doses reconcile across input, long and wide summaries.

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Topic 02: Data Summary

Data types · descriptive statistics · distribution · grouped summaries · basic plots and tables

Variable type determines valid summaries

- **Nominal:** Categories with no inherent order.
- **Ordinal:** Categories with an order but not equal intervals.
- **Discrete:** Countable numeric outcomes.
- **Continuous:** Measurements on an interval or ratio scale.

Qualitative and quantitative answer different questions

- **Qualitative:** What kinds, themes or experiences occur?
- **Quantitative:** How many, how much or how strongly?
- **Mixed evidence:** Counts locate patterns; text explains context and meaning.
- **Risk:** Do not turn labels into arithmetic merely by coding numbers.

Frequency tables reveal category structure

- **Count:** Number of observations in each level.
- **Proportion:** Count divided by a clearly stated denominator.
- **Missing:** Display unknown or excluded categories explicitly.
- **Comparison:** Use common denominators when comparing groups.

Mean describes balance; median describes position

- **Mean:** Uses every value and is sensitive to extremes.
- **Median:** Splits ordered observations into two halves.
- **Symmetry:** Mean and median are usually close.
- **Skew:** Their separation is a warning to inspect distribution.

Mean and median respond differently to outliers

- **Baseline:** Without extremes, centre measures often agree.
- **Outlier:** A large value pulls the mean more than the median.
- **Choice:** Select the statistic that answers the research question.
- **Disclosure:** Report distribution and sample size beside the centre.

Trimmed means trade sensitivity for robustness

- **Rule:** Remove a fixed proportion from each tail before averaging.
- **Benefit:** Reduces the influence of extreme observations.
- **Cost:** Discarded tails may contain real, important cases.
- **Use:** Compare trim levels rather than selecting one after the fact.

Range is simple but fragile

- **Definition:** Maximum minus minimum.
- **Strength:** Easy to calculate and explain.
- **Weakness:** Depends on only two observations.
- **Use:** Pair with IQR or standard deviation and the distribution.

Variance and standard deviation quantify spread

- **Variance:** Average squared deviation, estimated with $n-1$ for a sample.
- **SD:** Square root of variance in the original unit.
- **Meaning:** Larger SD means observations are more dispersed around the mean.
- **Limit:** Strong skew or outliers can make SD hard to interpret alone.

The IQR focuses on the middle half

- **Q1:** 25% of observations lie at or below this value.
- **Q3:** 75% lie at or below this value.
- **IQR:** Q3 − Q1, resistant to tail extremes.
- **Outlier rule:** 1.5 × IQR flags values for investigation, not automatic deletion.

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A five-number summary sketches the distribution

- **Minimum:** Lowest observed value.
- **Quartiles:** Q1, median and Q3 locate the middle structure.
- **Maximum:** Highest observed value.
- **Review:** Compare with units, n and possible data-entry errors.

Histograms depend on bin choices

- **Bins:** Intervals convert continuous values into counts.
- **Shape:** Reveal skew, modes, gaps and tails.
- **Sensitivity:** Different widths can hide or exaggerate patterns.
- **Control:** Use a defensible rule and show the underlying range.

Box plots make group comparison compact

- **Box:** Middle 50% from Q1 to Q3.
- **Line:** Median within the box.
- **Whiskers:** Typical range under the chosen convention.
- **Points:** Potential outliers requiring context and validation.

Grouped summaries require denominators

- **n:** Shows how much evidence supports each group statistic.
- **Missing:** State whether excluded cases differ by group.
- **Unequal groups:** Precision and visual weight differ with sample size.
- **Aggregation:** Do not compare sums where rates or means answer the question.

Sampling links observed data to a population

- **Population:** All units the research decision concerns.
- **Sample:** Observed subset used to estimate population characteristics.
- **Parameter:** Unknown population quantity.
- **Statistic:** Sample-based estimate that varies across samples.

Sampling distributions explain uncertainty

- **Repeated samples:** The same statistic takes different values.
- **Standard error:** Expected variability of an estimate across samples.
- **Sample size:** Larger n usually improves precision, not representativeness.
- **Bias:** Systematic selection errors remain even with more data.

Accuracy review precedes interpretation

- **Recompute:** Check a material statistic by an independent method.
- **Reconcile:** Counts and grouped totals agree with source data.
- **Inspect:** Plots reveal impossible values or unexpected shapes.
- **Document:** Record exclusions, rounding, units and package versions.

Summary tables should carry meaning

- **Title:** State population, measure and period.
- **Units:** Keep kg, cases, percent and counts explicit.
- **Precision:** Round only for communication; retain underlying values.
- **Footnote:** Explain missing data, denominators and method choices.

A defensible summary separates signal from caveat

- **Pattern:** What centre, spread or group difference is observed?
- **Magnitude:** How large is it in the original unit?
- **Uncertainty:** How much evidence and variability surround it?
- **Limit:** What should the reader not conclude from the summary?

Lab 3: Summarise Centre, Spread and Distribution

Alignment	LO2
R script	lab-03-descriptive-statistics.R

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Excel evidence	lab-03-descriptive-statistics.xlsx
Expected output	A descriptive-statistics table, distribution plot, outlier review and accuracy checklist.
Tools	R, mean, median, sd, IQR, quantile, hist, boxplot

Goal. Compare mean, trimmed mean and median, quantify variability and inspect how outliers affect health-service waiting-time summaries.

Before you begin

- Work from the numbered lab folder and keep its script, workbook and data subfolder together.
- Use the supplied synthetic/legacy training data only; it is not clinical evidence.
- Open the Excel workbook first and read the scenario, definitions and control expectations.
- Run the R script from the lab folder so relative data/ and outputs/ paths resolve correctly.
- Preserve the source CSV files. Put generated evidence only in outputs/ and the Excel Evidence sheet.

Detailed procedure

1	**Read the analytical question, variable definition and units in the workbook.** Open lab-03-descriptive-statistics.xlsx
2	**Run the R script and inspect missingness and the distribution before calculating summaries.** source('lab-03-descriptive-statistics.R', echo = TRUE)
3	**Compare mean, 10% and 20% trimmed means with the median.** Open outputs/lab-03-summary.csv
4	**Calculate range, variance, standard deviation, IQR and selected percentiles.** Review the metric definitions in the Learner Guide
5	**Identify possible outliers using the 1.5 × IQR rule without deleting them automatically.** Open outputs/lab-03-outliers.csv
6	**Inspect the histogram and box plot and explain skew or unusual values.** Open outputs/lab-03-distribution.png
7	**Recalculate one statistic independently in the Excel Control sheet.** Compare Excel check with R output
8	**State which measure of centre is most defensible and why.** Complete the Evidence sheet

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Troubleshooting

- File not found: confirm the script is running from the lab folder and the named CSV exists under data/.
- Package not installed: the supplied scripts use base-R fallbacks where possible; read the console message before installing anything.
- Unexpected factor or text values: inspect unique(), trim whitespace and compare with the workbook data dictionary.
- Plot file is blank or locked: close any open PNG, rerun the script and confirm dev.off() completed.
- Control does not reconcile: stop interpretation, identify the transformation stage where rows or totals diverged, and document the exception.

Evidence and acceptance

- Retain the runnable R script and full console output.
- Retain generated CSV/PNG/session evidence from outputs/.
- Complete the Excel Control, Evidence and Run Log sheets.
- State the finding, magnitude/effect, uncertainty, decision relevance and limitation in your own words.

R and Excel checks agree, units and sample size are stated, outliers remain traceable, and the chosen centre measure matches the observed distribution.

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Lab 4: Build Grouped Tables and Basic Plots

Alignment	LO2
R script	lab-04-grouped-summary.R
Excel evidence	lab-04-grouped-summary.xlsx
Expected output	Frequency/proportion tables, grouped numerical summaries, bar and box plots, and a denominator audit.
Tools	R, table, prop.table, aggregate, barplot, boxplot

Goal. Summarise health observations by category, preserve denominators and review the accuracy of proportions, grouped means and basic plots.

Before you begin

- Work from the numbered lab folder and keep its script, workbook and data subfolder together.
- Use the supplied synthetic/legacy training data only; it is not clinical evidence.
- Open the Excel workbook first and read the scenario, definitions and control expectations.
- Run the R script from the lab folder so relative data/ and outputs/ paths resolve correctly.
- Preserve the source CSV files. Put generated evidence only in outputs/ and the Excel Evidence sheet.

Detailed procedure

1	**Review categorical and numerical variable types in the workbook data dictionary.** Open lab-04-grouped-summary.xlsx
2	**Run the script and confirm levels, missing values and group sample sizes.** source('lab-04-grouped-summary.R', echo = TRUE)
3	**Create frequency and proportion tables for chest-pain category and smoking status.** Open outputs/lab-04-frequency-table.csv
4	**Calculate grouped means, medians and standard deviations for weight.** Open outputs/lab-04-grouped-statistics.csv
5	**Create a bar chart for counts and a box plot for weight by chest-pain category.** Open outputs/lab-04-grouped-plots.png
6	**Verify proportions sum to 1 within rounding tolerance and all groups state n.** Review outputs/lab-04-control.csv
7	**Compare the plot with the table and flag any interpretation not supported by both.** Complete Evidence!B6

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8	**Save the evidence workbook with one accurate group comparison and one limitation.** Complete the Evidence sheet

Troubleshooting

- File not found: confirm the script is running from the lab folder and the named CSV exists under data/.
- Package not installed: the supplied scripts use base-R fallbacks where possible; read the console message before installing anything.
- Unexpected factor or text values: inspect unique(), trim whitespace and compare with the workbook data dictionary.
- Plot file is blank or locked: close any open PNG, rerun the script and confirm dev.off() completed.
- Control does not reconcile: stop interpretation, identify the transformation stage where rows or totals diverged, and document the exception.

Evidence and acceptance

- Retain the runnable R script and full console output.
- Retain generated CSV/PNG/session evidence from outputs/.
- Complete the Excel Control, Evidence and Run Log sheets.
- State the finding, magnitude/effect, uncertainty, decision relevance and limitation in your own words.

Counts reconcile to the dataset, proportions sum to 100% within rounding, every grouped statistic includes n, and the written comparison matches both tables and plots.

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Topic 03: Quantitative Data Analysis

Correlation · regression · hypothesis testing · effect size · ANOVA · interpretation

Quantitative analysis begins with a model of the question

- **Outcome:** The response to explain, compare or predict.
- **Predictor:** A measured factor believed to relate to the outcome.
- **Design:** Paired, independent, observational or experimental.
- **Assumptions:** Conditions required for valid estimation or testing.

Covariance and correlation describe co-movement

- **Covariance:** Direction and joint variability in combined units.
- **Correlation:** Standardised direction and strength from −1 to +1.
- **Positive:** Variables tend to increase together.
- **Negative:** One tends to decrease as the other increases.

Correlation strength and direction

- **Near +1:** Strong positive linear association.
- **Near 0:** Little linear association, though nonlinear structure may exist.
- **Near −1:** Strong negative linear association.
- **Caution:** Outliers and restricted ranges can dominate the coefficient.

Pearson and rank correlation make different assumptions

- **Pearson:** Measures linear association between quantitative variables.
- **Spearman:** Measures monotonic association using ranks.
- **Choice:** Inspect form, scale and outliers before selecting.
- **Report:** Coefficient, interval, p-value, n and a scatter plot.

Correlation is not causation

- **Confounding:** A third factor may drive both variables.
- **Reverse direction:** The assumed predictor may actually respond to the outcome.
- **Selection:** Observed units may not represent the population.
- **Evidence:** Causal claims require design, timing and alternative explanations.

Linear regression expresses an average relationship

- **Intercept:** Expected outcome when predictors equal zero.
- **Slope:** Expected outcome change per unit of predictor.
- **Error:** Observed minus modelled outcome.
- **Fit:** Estimated line minimises squared residuals.

Regression coefficients need units and conditions

- **Direction:** Positive or negative adjusted relationship.
- **Magnitude:** Change in outcome units per predictor unit.
- **Condition:** Other included predictors held constant.
- **Limit:** Interpret within the observed data range.

R-squared measures explained variation

- **Range:** Usually 0 to 1 for models with an intercept.
- **Meaning:** Share of outcome variation explained in this sample.
- **Adjusted:** Penalises predictors that add little information.
- **Limit:** High fit does not prove causation or future accuracy.

Residuals reveal model mismatch

- **Randomness:** No systematic curve or pattern across fitted values.
- **Spread:** Variability should not fan out dramatically.
- **Outliers:** Large residuals require context and sensitivity review.
- **Dependence:** Time or clusters can violate independence.

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Hypothesis testing is a pre-specified decision rule

- **H0:** Baseline model or no-effect statement.
- **H1:** Difference or direction the study is designed to detect.
- **Alpha:** Chosen false-positive risk threshold.
- **Evidence:** Test statistic and p-value under H0 assumptions.

P-values have a narrow meaning

- **Conditional:** Calculated assuming H0 and the test assumptions.
- **Extremeness:** Probability of data at least this extreme under H0.
- **Not effect:** Does not tell how large or useful the finding is.
- **Not truth:** Is not the probability that H0 is true.

One- and two-tailed tests answer different questions

- **Two-tailed:** Detects differences in either direction.
- **One-tailed:** Only a justified pre-specified direction counts.
- **Integrity:** Do not choose the tail after observing the result.
- **Risk:** The opposite direction may still matter operationally.

Type I and Type II errors trade off

- **Type I:** Reject a true H0: false positive.
- **Type II:** Fail to reject a false H0: false negative.
- **Power:** Probability of detecting a meaningful true effect.
- **Design:** Sample size, variability and effect magnitude affect both.

T-tests must match how data was collected

- **One sample:** Compare a sample mean with a benchmark.
- **Independent:** Compare two separate groups.
- **Paired:** Compare matched units or repeated measures.
- **Welch:** Allows unequal group variances and is often safer.

Effect size complements significance

- **Mean difference:** Magnitude in the original unit.
- **Standardised:** Difference scaled by variability.
- **Interval:** Range of plausible effect values under assumptions.
- **Decision:** Compare with cost, risk and practical threshold.

ANOVA separates within and between variation

- **Between:** Variation explained by differences among group means.
- **Within:** Variation among observations inside groups.
- **F ratio:** Between-group signal relative to within-group noise.
- **Follow-up:** Post-hoc comparisons identify which groups differ.

Assumptions are evidence, not a checkbox

- **Independence:** Comes primarily from study design.
- **Distribution:** Inspect residuals or within-group shapes.
- **Variance:** Review group spreads; use robust alternatives when needed.
- **Sensitivity:** Check whether conclusions change under defensible choices.

Interpretation is a structured argument

- **Estimate:** State direction and magnitude.
- **Uncertainty:** Give interval, variability and sample size.
- **Evidence:** Report test/model and p-value without overclaim.
- **Limit:** Name assumptions, design constraints and next evidence.

Lab 5: Analyse Correlation and Regression

Alignment	LO3
R script	lab-05-correlation-regression.R

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Excel evidence	lab-05-correlation-regression.xlsx
Expected output	Correlation evidence, fitted regression model, diagnostic plot, prediction and limitations statement.
Tools	R, cor, cor.test, lm, summary, predict, residual diagnostics

Goal. Examine a quantitative relationship, fit a linear model, evaluate residuals and communicate an interpretation without causal overclaiming.

Before you begin

- Work from the numbered lab folder and keep its script, workbook and data subfolder together.
- Use the supplied synthetic/legacy training data only; it is not clinical evidence.
- Open the Excel workbook first and read the scenario, definitions and control expectations.
- Run the R script from the lab folder so relative data/ and outputs/ paths resolve correctly.
- Preserve the source CSV files. Put generated evidence only in outputs/ and the Excel Evidence sheet.

Detailed procedure

1	**Define the outcome, predictor, population and analytical question before running a model.** Record the model design in the workbook
2	**Run the script and inspect the scatter plot for direction, form and unusual observations.** source('lab-05-correlation-regression.R', echo = TRUE)
3	**Calculate Pearson correlation with a confidence interval and p-value.** Open outputs/lab-05-correlation.csv
4	**Fit the linear model and extract coefficients, R-squared and model significance.** Open outputs/lab-05-model-summary.csv
5	**Interpret the slope in units and distinguish association from causation.** Complete Evidence!B4:B5
6	**Review residual-versus-fitted and normal-Q-Q plots for visible problems.** Open outputs/lab-05-model-diagnostics.png
7	**Generate one within-range prediction with an interval.** Open outputs/lab-05-prediction.csv
8	**State one strength and two limitations of the technique.** Complete the reflection

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Troubleshooting

- File not found: confirm the script is running from the lab folder and the named CSV exists under data/.
- Package not installed: the supplied scripts use base-R fallbacks where possible; read the console message before installing anything.
- Unexpected factor or text values: inspect unique(), trim whitespace and compare with the workbook data dictionary.
- Plot file is blank or locked: close any open PNG, rerun the script and confirm dev.off() completed.
- Control does not reconcile: stop interpretation, identify the transformation stage where rows or totals diverged, and document the exception.

Evidence and acceptance

- Retain the runnable R script and full console output.
- Retain generated CSV/PNG/session evidence from outputs/.
- Complete the Excel Control, Evidence and Run Log sheets.
- State the finding, magnitude/effect, uncertainty, decision relevance and limitation in your own words.

The model output is reproducible, coefficient and R-squared interpretations use correct units, diagnostics are reviewed, and the conclusion avoids causal claims.

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Lab 6: Test Differences and Compare Groups

Alignment	LO3
R script	lab-06-hypothesis-anova.R
Excel evidence	lab-06-hypothesis-anova.xlsx
Expected output	Test-design record, t-test and ANOVA outputs, effect estimates, assumption checks and conclusion.
Tools	R, t.test, aov, TukeyHSD, effect-size calculation, boxplot

Goal. Pre-specify hypotheses, run a two-sample t-test and one-way ANOVA, quantify effects and separate statistical from practical significance.

Before you begin

- Work from the numbered lab folder and keep its script, workbook and data subfolder together.
- Use the supplied synthetic/legacy training data only; it is not clinical evidence.
- Open the Excel workbook first and read the scenario, definitions and control expectations.
- Run the R script from the lab folder so relative data/ and outputs/ paths resolve correctly.
- Preserve the source CSV files. Put generated evidence only in outputs/ and the Excel Evidence sheet.

Detailed procedure

1	**Write the research questions, H0/H1, tail and alpha before viewing test results.** Complete the Design sheet
2	**Run the script and inspect group sizes, missingness, outliers and distributions.** source('lab-06-hypothesis-anova.R', echo = TRUE)
3	**Run Welch's two-sample t-test for weight by smoking status.** Open outputs/lab-06-t-test.csv
4	**Calculate the mean difference, confidence interval and standardised effect.** Review outputs/lab-06-effect-size.csv
5	**Run one-way ANOVA for cancer-cell count across chest-pain categories.** Open outputs/lab-06-anova.csv
6	**If the omnibus result is significant, review Tukey pairwise comparisons.** Open outputs/lab-06-tukey.csv
7	**Compare p-values with alpha and explain Type I/II risk and practical meaning.** Complete the Evidence sheet

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8	**Record limitations and the next evidence required for a stronger conclusion.** Complete Evidence!B9

Troubleshooting

- File not found: confirm the script is running from the lab folder and the named CSV exists under data/.
- Package not installed: the supplied scripts use base-R fallbacks where possible; read the console message before installing anything.
- Unexpected factor or text values: inspect unique(), trim whitespace and compare with the workbook data dictionary.
- Plot file is blank or locked: close any open PNG, rerun the script and confirm dev.off() completed.
- Control does not reconcile: stop interpretation, identify the transformation stage where rows or totals diverged, and document the exception.

Evidence and acceptance

- Retain the runnable R script and full console output.
- Retain generated CSV/PNG/session evidence from outputs/.
- Complete the Excel Control, Evidence and Run Log sheets.
- State the finding, magnitude/effect, uncertainty, decision relevance and limitation in your own words.

Test choices match the designs, hypotheses were pre-specified, effect estimates accompany p-values, and conclusions state assumptions and practical limitations.

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Topic 04: Qualitative Data Analysis

Text preparation · tokenisation · stop words · term frequency · word clouds · thematic interpretation

Qualitative data preserves experience and context

- **Open responses:** Capture language respondents choose themselves.
- **Themes:** Recurring meanings can be coded and compared.
- **Context:** The same word can carry different meanings in different excerpts.
- **Responsibility:** Protect identity and avoid decontextualised quotation.

A qualitative workflow remains auditable

- **Preserve:** Keep original response and stable record ID.
- **Prepare:** Create derived cleaned text without overwriting source.
- **Code:** Apply documented tokens, themes or categories.
- **Review:** Trace every claim back to representative evidence.

Text normalisation changes the evidence

- **Case:** Lowercase merges capitalisation variants.
- **Punctuation:** Removal can change contractions, hashtags or medical codes.
- **Spelling:** Correction can merge meaning or introduce analyst judgement.
- **Audit:** Record each rule and retain the original text.

Tokenisation defines the analytical unit

- **Word:** Simple and interpretable, but loses phrase meaning.
- **N-gram:** Preserves short phrases and negation better.
- **Sentence:** Supports context and sentiment review.
- **Theme:** Human-coded concept may be more meaningful than a token.

Stop words reduce noise—and can remove meaning

- **General:** Common function words often add little topical information.
- **Domain:** Frequently used terms may still be important in healthcare.
- **Negation:** Removing not or no can reverse meaning.
- **Sensitivity:** Compare results before and after the list changes.

Term frequency is a starting point

- **Count:** How often a token appears.
- **Document frequency:** How many responses contain the token.
- **Rate:** Count relative to total words or responses.
- **Limit:** Frequency alone does not establish sentiment, severity or importance.

Term frequency changes after stop-word control

- **Raw:** High-frequency grammar terms dominate.
- **General stop list:** Topic words become more visible.
- **Domain stop list:** Prominence can change materially.
- **Evidence:** Publish the ranked table beside the visual.

Concordance restores local context

- **Keyword:** Choose a term suggested by frequency or theory.
- **Window:** Inspect neighbouring words or full response.
- **Meaning:** Separate different uses of the same term.
- **Trace:** Retain record ID without exposing personal identity.

A codebook turns interpretation into a process

- **Code:** Short label for a defined meaning.
- **Definition:** What is included and excluded.
- **Example:** Representative excerpt illustrating the code.
- **Decision rule:** How ambiguous cases are resolved.

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Human coding needs consistency checks

- **Calibration:** Coders discuss examples before full coding.
- **Double-code:** Independent review reveals disagreements.
- **Resolution:** Record how differences were reconciled.
- **Reflexivity:** Acknowledge how analyst perspective shapes interpretation.

Word clouds are overview visuals, not analysis

- **Size:** Usually encodes term frequency.
- **Strength:** Rapidly surfaces prominent vocabulary.
- **Weakness:** Poor for precise comparison and hides context.
- **Companion:** Always provide a ranked table and excerpts.

A reproducible word cloud fixes random choices

- **Seed:** Keeps the layout stable across reruns.
- **Threshold:** State minimum frequency and maximum words.
- **Palette:** Use readable, accessible colours.
- **Input:** Archive the exact ranked term table.

Preprocessing sensitivity is an effect

- **Choice:** Stop words, stemming and token rules change counts.
- **Compare:** Measure rank and frequency shifts.
- **Interpret:** Claims robust to sensible choices are more credible.
- **Report:** Disclose choices with material impact.

Stemming and lemmatisation trade detail for grouping

- **Stem:** Cuts words to a common root, sometimes awkwardly.
- **Lemma:** Maps variants to a dictionary form.
- **Benefit:** Combines related grammatical forms.
- **Risk:** Can merge meanings that should remain distinct.

Qualitative significance is not only a p-value

- **Salience:** Prominence within participant accounts.
- **Coverage:** Breadth across responses or channels.
- **Contrast:** Meaningful difference between contexts.
- **Impact:** Potential consequence supported by evidence and judgement.

Triangulation strengthens interpretation

- **Counts:** Show prevalence or distribution.
- **Excerpts:** Show meaning and context.
- **Groups:** Compare channels, time or participant categories.
- **Peer review:** Challenge selective reading and alternative explanations.

Qualitative visuals need ethical boundaries

- **Privacy:** Remove identifiable or sensitive details.
- **Representation:** Do not let vivid quotes stand in for all respondents.
- **Proportionality:** Avoid implying population prevalence from convenience feedback.
- **Purpose:** Use only for the stated research or service-improvement need.

A credible theme statement carries evidence and restraint

- **Theme:** Name the recurring idea precisely.
- **Support:** Provide counts, coverage and representative excerpts.
- **Variation:** Show exceptions or contrasting accounts.
- **Limit:** State sampling, coding and contextual constraints.

Lab 7: Prepare Text and Analyse Term Frequency

Alignment	LO4
R script	lab-07-text-analysis.R

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Excel evidence	lab-07-text-analysis.xlsx
Expected output	Clean token table, stop-word audit, frequency table, channel comparison and interpretation notes.
Tools	R, regular expressions, strsplit, stop-word filtering, table, aggregate

Goal. Convert feedback text into a traceable token table, remove documented stop words and compare term frequencies across service channels.

Before you begin

- Work from the numbered lab folder and keep its script, workbook and data subfolder together.
- Use the supplied synthetic/legacy training data only; it is not clinical evidence.
- Open the Excel workbook first and read the scenario, definitions and control expectations.
- Run the R script from the lab folder so relative data/ and outputs/ paths resolve correctly.
- Preserve the source CSV files. Put generated evidence only in outputs/ and the Excel Evidence sheet.

Detailed procedure

1	**Read the qualitative question, source context and coding limitations in the workbook.** Open lab-07-text-analysis.xlsx
2	**Run the script and preserve an unmodified copy of each response with its record ID.** source('lab-07-text-analysis.R', echo = TRUE)
3	**Normalise case and punctuation and split responses into word tokens.** Inspect outputs/lab-07-tokens.csv
4	**Remove blank tokens, numbers and the documented stop-word list.** Open outputs/lab-07-stopword-audit.csv
5	**Count terms overall and by feedback channel.** Open outputs/lab-07-term-frequency.csv
6	**Review high-frequency words in their original response context.** Use record_id to trace tokens back to data/feedback.csv
7	**Compare two channels without equating frequency with importance.** Complete Evidence!B5:B7
8	**Record one coding limitation and one peer-review check.** Complete the Evidence sheet

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Troubleshooting

- File not found: confirm the script is running from the lab folder and the named CSV exists under data/.
- Package not installed: the supplied scripts use base-R fallbacks where possible; read the console message before installing anything.
- Unexpected factor or text values: inspect unique(), trim whitespace and compare with the workbook data dictionary.
- Plot file is blank or locked: close any open PNG, rerun the script and confirm dev.off() completed.
- Control does not reconcile: stop interpretation, identify the transformation stage where rows or totals diverged, and document the exception.

Evidence and acceptance

- Retain the runnable R script and full console output.
- Retain generated CSV/PNG/session evidence from outputs/.
- Complete the Excel Control, Evidence and Run Log sheets.
- State the finding, magnitude/effect, uncertainty, decision relevance and limitation in your own words.

Every token remains traceable to a source record, exclusions are documented, frequency totals reconcile, and the interpretation uses context rather than frequency alone.

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Lab 8: Create and Critique a Word Cloud

Alignment	LO4
R script	lab-08-word-cloud.R
Excel evidence	lab-08-word-cloud.xlsx
Expected output	Word-cloud image, ranked-frequency table, sensitivity comparison, excerpt evidence and limitations statement.
Tools	R, wordcloud optional package, base-graphics fallback, RColorBrewer optional

Goal. Create a reproducible word cloud or frequency display, test the effect of preprocessing choices and triangulate the visual with representative excerpts.

Before you begin

- Work from the numbered lab folder and keep its script, workbook and data subfolder together.
- Use the supplied synthetic/legacy training data only; it is not clinical evidence.
- Open the Excel workbook first and read the scenario, definitions and control expectations.
- Run the R script from the lab folder so relative data/ and outputs/ paths resolve correctly.
- Preserve the source CSV files. Put generated evidence only in outputs/ and the Excel Evidence sheet.

Detailed procedure

1	**Review the term-frequency input and the visual question in the workbook.** Open lab-08-word-cloud.xlsx
2	**Run the script; it uses wordcloud when installed and a deterministic base-R layout otherwise.** source('lab-08-word-cloud.R', echo = TRUE)
3	**Create the ranked frequency table and retain the preprocessing settings.** Open outputs/lab-08-ranked-terms.csv
4	**Generate the word cloud with a fixed random seed and accessible palette.** Open outputs/lab-08-word-cloud.png
5	**Repeat with the domain stop-word list changed and compare the top terms.** Open outputs/lab-08-sensitivity.csv
6	**Trace three prominent terms back to representative source excerpts.** Complete the Excerpts sheet
7	**State what the cloud supports and what it cannot establish.** Complete Evidence!B7:B8

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8	**Record an effect or salience comparison using counts/proportions, not font size alone.** Complete the Evidence sheet

Troubleshooting

- File not found: confirm the script is running from the lab folder and the named CSV exists under data/.
- Package not installed: the supplied scripts use base-R fallbacks where possible; read the console message before installing anything.
- Unexpected factor or text values: inspect unique(), trim whitespace and compare with the workbook data dictionary.
- Plot file is blank or locked: close any open PNG, rerun the script and confirm dev.off() completed.
- Control does not reconcile: stop interpretation, identify the transformation stage where rows or totals diverged, and document the exception.

Evidence and acceptance

- Retain the runnable R script and full console output.
- Retain generated CSV/PNG/session evidence from outputs/.
- Complete the Excel Control, Evidence and Run Log sheets.
- State the finding, magnitude/effect, uncertainty, decision relevance and limitation in your own words.

The visual is reproducible, ranked counts are available, preprocessing sensitivity is documented, and every claim is supported by counts plus traceable excerpts.

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Topic 05: Data Visualization

Grammar of graphics · quantitative and qualitative plots · customisation · accessibility · findings

A visualisation answers a question

- **Trend:** How does a measure change over ordered time?
- **Comparison:** Which categories are higher, lower or different?
- **Distribution:** How are values spread and where are unusual observations?
- **Relationship:** How do two quantitative variables move together?

The grammar of graphics separates decisions

- **Data:** The analytical table and variables.
- **Aesthetics:** Mappings to x, y, colour, size, shape or group.
- **Geom:** Marks such as point, line, bar or box.
- **Layers:** Statistics, scales, coordinates, facets and annotation.

Data and aesthetics define the visual contract

- **x/y:** Position is usually the most precise quantitative encoding.
- **Colour:** Useful for grouping but weak without redundant cues.
- **Size:** Area judgements are less precise than position or length.
- **Group:** Controls which observations connect or aggregate together.

Geoms must match the observation and question

- **Point:** One mark per paired quantitative observation.
- **Line:** Connects ordered observations in a meaningful sequence.
- **Bar:** Compares category values using aligned lengths.
- **Box:** Summarises a quantitative distribution by group.

Statistical transforms create derived displays

- **Count:** Bars can count records by category.
- **Bin:** Histograms group continuous values into intervals.
- **Smooth:** A fitted trend summarises relationship form.
- **Summary:** Intervals and point estimates reduce data visibly.

Scales translate data into perception

- **Domain:** Values included in the mapping.
- **Range:** Positions, colours or sizes used on screen.
- **Breaks:** Tick marks guide interpretation.
- **Labels:** Units and formats make values legible.

Trend visual: signal over time

- **Order:** Use a true date or other meaningful sequence.
- **Continuity:** Connect only observations that belong together.
- **Context:** Show period, unit and notable events.
- **Uncertainty:** Add intervals or caveats where estimates are noisy.

Comparison visual: zero baseline matters

- **Bars:** Length should start at zero for honest magnitude comparison.
- **Sort:** Order categories to reveal structure.
- **Direct labels:** Reduce lookup effort where values matter.
- **Limit:** Avoid 3D effects and decorative area distortion.

Distribution visuals reveal what averages hide

- **Histogram:** Shape across numeric intervals.
- **Box plot:** Compact group comparison of median, IQR and outliers.
- **Density:** Smoothed shape sensitive to bandwidth.
- **Raw points:** Show sample size and clusters where feasible.

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Quantitative relationship: scatter plus trend

- **Points:** Preserve individual observations.
- **Trend:** Summarise form without hiding dispersion.
- **Groups:** Colour, shape or facets expose structure.
- **Caution:** A fitted line communicates association, not causal proof.

Qualitative plots need precise companions

- **Ranked bars:** Best for comparing term or theme frequencies.
- **Word cloud:** Useful only as an overview of prominence.
- **Heat map:** Shows frequency across groups or time.
- **Excerpts:** Restore meaning behind counts and colours.

Facets create small multiples

- **Consistency:** Same axes and visual grammar across panels.
- **Comparison:** Group patterns become visible without overplotting.
- **Scale:** Use free scales only when comparison is not the goal.
- **Order:** Arrange panels in a meaningful sequence.

Colour should encode, not decorate

- **Categorical:** Distinct hues for unordered groups.
- **Sequential:** Light-to-dark progression for ordered magnitude.
- **Diverging:** Two directions around a meaningful midpoint.
- **Accessibility:** Check contrast and add labels, shapes or facets.

Annotation guides attention with evidence

- **Direct:** Label the important point or change.
- **Reference:** Add target, benchmark or event line.
- **Restraint:** Annotate the finding, not every mark.
- **Source:** Keep period, population and unit visible.

Themes improve consistency without hiding data

- **Hierarchy:** Title, plot, annotation and source have distinct weight.
- **Typography:** Readable sizes and few font styles.
- **Grid:** Use light guides only where they help estimation.
- **Whitespace:** Separate elements and reduce cognitive load.

Misleading visuals change the argument

- **Truncated bar axis:** Exaggerates small category differences.
- **Dual axes:** Can manufacture apparent relationships.
- **3D:** Distorts position, length and area.
- **Cherry-picking:** Selective periods or groups hide contradictory evidence.

Visual QA is an analytical control

- **Reconcile:** Displayed values agree with the source summary.
- **Inspect:** No clipping, overlap, wrong labels or hidden categories.
- **Challenge:** Test alternative scale, grouping or period choices.
- **Accessibility:** Meaning survives greyscale and colour-vision differences.

Interpret findings in five parts

- **Pattern:** What is visibly happening?
- **Magnitude:** How large is it and in what unit?
- **Uncertainty:** What variability or evidence limits confidence?
- **Relevance:** Why does it matter to the research decision?
- **Limit:** What conclusion is not supported?

Lab 9: Build Layered Quantitative Visuals

Alignment	LO5

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R script	lab-09-ggplot-visuals.R
Excel evidence	lab-09-ggplot-visuals.xlsx
Expected output	A bar chart, faceted distribution plot and annotated scatter plot with a chart-choice rationale.
Tools	R, ggplot2 when available, base-R fallback, scales, facets, annotation

Goal. Use the grammar of graphics to create comparison, distribution and relationship plots with truthful scales, useful labels and accessible emphasis.

Before you begin

- Work from the numbered lab folder and keep its script, workbook and data subfolder together.
- Use the supplied synthetic/legacy training data only; it is not clinical evidence.
- Open the Excel workbook first and read the scenario, definitions and control expectations.
- Run the R script from the lab folder so relative data/ and outputs/ paths resolve correctly.
- Preserve the source CSV files. Put generated evidence only in outputs/ and the Excel Evidence sheet.

Detailed procedure

1	**Translate each analytical question into a chart purpose and required variables.** Complete the Chart Plan sheet
2	**Run the script and inspect the data types and ranges before mapping aesthetics.** source('lab-09-ggplot-visuals.R', echo = TRUE)
3	**Create a sorted bar chart for a categorical comparison with a zero baseline.** Open outputs/lab-09-comparison.png
4	**Create a distribution view across meaningful groups.** Open outputs/lab-09-distribution.png
5	**Create a scatter plot with colour/faceting and an appropriate fitted trend.** Open outputs/lab-09-relationship.png
6	**Improve labels, units, theme, contrast and redundant encodings.** Review the Accessibility checklist
7	**Check that no scale, aspect ratio or annotation exaggerates the evidence.** Complete the Truthfulness audit
8	**Write one pattern, one uncertainty and one limitation for each plot.** Complete the Evidence sheet

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Troubleshooting

- File not found: confirm the script is running from the lab folder and the named CSV exists under data/.
- Package not installed: the supplied scripts use base-R fallbacks where possible; read the console message before installing anything.
- Unexpected factor or text values: inspect unique(), trim whitespace and compare with the workbook data dictionary.
- Plot file is blank or locked: close any open PNG, rerun the script and confirm dev.off() completed.
- Control does not reconcile: stop interpretation, identify the transformation stage where rows or totals diverged, and document the exception.

Evidence and acceptance

- Retain the runnable R script and full console output.
- Retain generated CSV/PNG/session evidence from outputs/.
- Complete the Excel Control, Evidence and Run Log sheets.
- State the finding, magnitude/effect, uncertainty, decision relevance and limitation in your own words.

All three plots answer their stated questions, remain legible without colour alone, use truthful scales and carry interpretations supported by the displayed data.

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Lab 10: Capstone: Review and Communicate Findings

Alignment	LO5
R script	lab-10-capstone.R
Excel evidence	lab-10-capstone.xlsx
Expected output	An auditable analysis bundle with control log, statistical results, three decision visuals and a concise evidence-led recommendation.
Tools	R, base statistics, optional ggplot2, visual QA, reproducibility and interpretation

Goal. Integrate preparation, summary, modelling, text evidence and visualisation into a compact reproducible healthcare research briefing.

Before you begin

- Work from the numbered lab folder and keep its script, workbook and data subfolder together.
- Use the supplied synthetic/legacy training data only; it is not clinical evidence.
- Open the Excel workbook first and read the scenario, definitions and control expectations.
- Run the R script from the lab folder so relative data/ and outputs/ paths resolve correctly.
- Preserve the source CSV files. Put generated evidence only in outputs/ and the Excel Evidence sheet.

Detailed procedure

1	**Read the capstone decision question, audience and acceptance criteria.** Open lab-10-capstone.xlsx
2	**Run the end-to-end script and review the generated session and control logs.** source('lab-10-capstone.R', echo = TRUE)
3	**Verify row counts, missing values, factor levels and material totals before analysis.** Open outputs/lab-10-control-log.csv
4	**Review the descriptive summary, model result and qualitative frequency evidence.** Open outputs/lab-10-analysis-summary.csv
5	**Inspect the three decision visuals and confirm labels, units, scales and source period.** Open outputs/lab-10-briefing.png
6	**Challenge the first interpretation using an alternative grouping or sensitivity check.** Open outputs/lab-10-sensitivity.csv
7	**Write the evidence statement: pattern, magnitude, uncertainty, relevance and limitation.** Complete Evidence!B4:B8

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8	**Run the peer-review checklist and record release owner, version and date.** Complete Release Checklist and save the workbook

Troubleshooting

- File not found: confirm the script is running from the lab folder and the named CSV exists under data/.
- Package not installed: the supplied scripts use base-R fallbacks where possible; read the console message before installing anything.
- Unexpected factor or text values: inspect unique(), trim whitespace and compare with the workbook data dictionary.
- Plot file is blank or locked: close any open PNG, rerun the script and confirm dev.off() completed.
- Control does not reconcile: stop interpretation, identify the transformation stage where rows or totals diverged, and document the exception.

Evidence and acceptance

- Retain the runnable R script and full console output.
- Retain generated CSV/PNG/session evidence from outputs/.
- Complete the Excel Control, Evidence and Run Log sheets.
- State the finding, magnitude/effect, uncertainty, decision relevance and limitation in your own words.

The bundle reruns from source data, all controls pass or are explained, visuals agree with statistical outputs, and the recommendation distinguishes evidence from judgement.

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